

1. (d)  $\vec{r} = 5t^2\hat{i} - 5t^2\hat{j}$   
Velocity is given by  
 $\frac{d\vec{r}}{dt} = \vec{v} = 10t\hat{i} - 10t\hat{j}$   
At  $t = 2$  sec  
 $\vec{v} = 20\hat{i} - 20\hat{j}$ ,  $\tan\theta = \frac{v_y}{v_x} = \frac{-20}{20} = -1$

From -ve Y-axis  
 $\theta = \tan^{-1} 1$

(d)  $\vec{v} = \frac{d\vec{r}}{dt} = 4t\hat{i}$   
Hence, velocity of ant at  $t = 1$  sec  
 $\vec{v} = 4\hat{i}$

3. [673] Given:  $a_1 = 3 \text{ m/s}^2$ ,  $u_1 = 5 \text{ m/s}$   
 $x = 84 \text{ m}$   
 $v_1^2 - u_1^2 = 2ax$   
 $v_1^2 - 25 = 2(3)(84) \Rightarrow v_1 = 23 \text{ m/s}$   
Now,  $v_1 - u_2 = at \Rightarrow t = \frac{23-5}{3} = 6 \text{ s}$

$v_1 = 0 - a_2 t = 0 - 2 \times (6) = 12 \text{ m/s}$   
 $v^2 = v_1^2 + v_2^2 = 23^2 + 12^2 = 673$   
 $v = \sqrt{673} \text{ m/s}$

4. (d) Given:  $x = 2 - 4t$  ... (i)  
 $\frac{dx}{dt} = v_x = 4$  and  $\frac{dy}{dt} = v_y = 0$   
Now,  $y = 3t + 8t^2$  ... (ii)

$\frac{dy}{dt} = v_y = 3 + 16t$  and  
 $\frac{dv_y}{dt} = a_y = 16$

Hence, the motion will be uniformly accelerated motion.

From path,  $x = 2 - 4t$   
 $\Rightarrow t = \frac{x-2}{-4}$

Put this value of  $t$  in equation (ii)

$y = 3\left(\frac{x-2}{-4}\right) + 8\left(\frac{x-2}{-4}\right)^2$

5. [2]  $\vec{a} = \frac{\vec{F}}{m} = 100\hat{i} + 50\hat{j}$

$\vec{s} = \vec{ut} + \frac{1}{2}\vec{at}^2$

$a\hat{i} + b\hat{j} = 0 + \frac{1}{2}(100\hat{i} + 50\hat{j}) \times 4$

$= 200\hat{i} + 100\hat{j} \Rightarrow \frac{a}{b} = \frac{200}{100} = 2$

6. (b) We have,  $x = 3t$ ,  $y = 5t^2$   
 $v_x = \frac{dx}{dt} = 3 \Rightarrow v_y = \frac{dy}{dt} = 10t$   
At  $t = 2$  sec

$a_x = \frac{dv_x}{dt} = 0 \Rightarrow a_y = \frac{dv_y}{dt} = 30t$

$\vec{a} = a_x\hat{i} + a_y\hat{j} = a_y\hat{j}$

Since, z coordinate is constant. Therefore,  $a_z = 0$

$\vec{a} = 30\hat{j}$

at  $t = 1$  sec  $\vec{a} = 30\hat{j}$

7. [580]  $X'_y = -3t^2 + 8t + 10$

$\vec{v}_1 = (-6t + 8)\hat{j} = 2\hat{j}$

$\vec{v}_2 = 5 - 8t^2$

$\vec{v}_3 = -24t^2\hat{j}$

$\sqrt{v} = |\vec{v}_1 - \vec{v}_2| = |-24\hat{j} - 2\hat{j}|$

$\sqrt{v} = \sqrt{24^2 + 2^2}$

$v = 580 \text{ m/s}$

8. (d)  $S_y = u_y t + \frac{1}{2}a_y t^2$   
 $32 = 0 + \frac{1}{2} \times 4t^2 \Rightarrow t = 4 \text{ sec}$

$S_x = u_x t + \frac{1}{2}a_x t^2 = 3 \times 4 + \frac{1}{2} \times 6 \times 16 = 60 \text{ m}$

9. (d) At  $t = t$ ,  $(20 \text{ m}, y_0)$   
 $u = 5\hat{j} \text{ m/s}$   
 $a_x = 10\hat{i}$   
 $a_y = 4\hat{j}$   
 $x = \frac{1}{2}a_x t^2$

$\therefore t = \sqrt{\frac{20 \times 2}{10}} = 2 \text{ s}$

$\therefore y_0 = 5 \times 2 + \frac{1}{2} \times 4 \times 2^2 = 18 \text{ m}$

10. (a)  $\frac{h_1}{d} = \tan 45^\circ$

$\frac{h_1 + h_2}{d + 2.464d} = \tan 30^\circ \Rightarrow h_2 = d$

11. (c)  $\frac{dx}{dt} = kv_y$ ,  $\frac{dy}{dt} = kx$

$\frac{dx}{dy} = \frac{y}{x} \Rightarrow x^2 = y^2 + c$

$\frac{dx}{dy} = \frac{y}{x}$

12. (b)  $\vec{s} = (5\hat{i} + 4\hat{j})t + \frac{1}{2}(4\hat{i} + 4\hat{j})t^2$

$= 10\hat{i} + 8\hat{j} + 2\hat{i}t^2 + 2\hat{j}t^2$

$\Rightarrow \vec{r}_2 - \vec{r}_1 = 18\hat{i} + 16\hat{j}$

$\vec{r} = 20\hat{i} + 20\hat{j} \Rightarrow |\vec{r}| = 20\sqrt{2}$

13. (a)  $v_x = \frac{dx}{dt} = -a \sin \omega t$   
 $v_y = \frac{dy}{dt} = a \cos \omega t \Rightarrow v_z = \frac{dz}{dt} = a \omega$

$\therefore v = \sqrt{v_x^2 + v_y^2 + v_z^2} = a\omega\sqrt{2}$

14. (d)  $\vec{r} = 15t^2\hat{i} + (4 - 20t^2)\hat{j}$   
 $\vec{v} = \frac{d\vec{r}}{dt} = 30t\hat{i} + (-40t)\hat{j}$

$\vec{a} = \frac{d\vec{v}}{dt} = 30\hat{i} - 40\hat{j}$

$|\vec{a}| = 50 \text{ m/s}^2$

15. (d) Range =  $3H_{\max}$  (given)  
 $\Rightarrow u^2 \sin 2\theta / g = 3(u^2 \sin^2 \theta / 2g)$

$2 \sin \theta \cos \theta = 3/2 \sin^2 \theta$

$\tan \theta = 4/3 \Rightarrow \theta = 53^\circ$

$R = u^2 (2 \times 3/5 \times 4/5) / g = 24u^2 / 25g$

16. (c) The maximum height  $h_m$  of a projectile, with projection angle  $\phi$  measured from the vertical, is given by:

$h_m = \frac{u^2 \cos^2 \phi}{2g}$

From the diagrams, option (c) correctly represents the trajectory where  $\phi$  from the vertical corresponds to a parabolic path with the given height function. Thus, the answer is (c).

17. (a) The maximum height of the first ball is 8 times that of the second ball:

$(H_{\max})_1 = 8 \times (H_{\max})_2$

Maximum Height Formula:

$H_{\max} = \frac{u^2 \sin^2 \theta}{2g}$

Therefore,

$\frac{u^2 \sin^2 \theta_1}{2g} = 8 \times \frac{u^2 \sin^2 \theta_2}{2g}$

$\sin^2 \theta_1 = 8 \sin^2 \theta_2 \Rightarrow \sin \theta_1 = 2\sqrt{2} \sin \theta_2$

Time of Flight

$T = \frac{2u \sin \theta}{g}$

Taking the ratio of the times of flight for the two balls:

$\frac{T_1}{T_2} = \frac{\frac{2u \sin \theta_1}{g}}{\frac{2u \sin \theta_2}{g}} = \frac{\sin \theta_1}{\sin \theta_2} = 2\sqrt{2}$

18. (d) Time of flight,  $T = \frac{2u \sin \theta}{g}$

Put  $\theta_1 = 45^\circ$ ;  $\theta_2 = 45^\circ - \alpha$

$\frac{T_1}{T_2} = \frac{\sin(45^\circ + \alpha)}{\sin(45^\circ - \alpha)}$

$\sin(a+b) = \sin a \cos b + \cos a \sin b$

$\sin(a-b) = \sin a \cos b - \cos a \sin b$

$\frac{T_1}{T_2} = \frac{\frac{1}{\sqrt{2}} \times \cos \alpha + \frac{1}{\sqrt{2}} \times \sin \alpha}{\frac{1}{\sqrt{2}} \times \cos \alpha - \frac{1}{\sqrt{2}} \times \sin \alpha}$

$\Rightarrow \frac{T_1}{T_2} = \frac{\cos \alpha + \sin \alpha}{\cos \alpha - \sin \alpha} = \frac{1 + \tan \alpha}{1 - \tan \alpha}$

19. [5]  $u_y = 60 \sin 30^\circ$  (30 m/s)  
Height attained in the first second,

$h_0 = (u_y) \cdot 1 - \frac{1}{2}g(1)^2$

$h_0 = 30 - 5 = 25 \text{ m}$

$h_1 = 30 + \left(-\frac{10}{2}\right) \times (2 \times 3 - 1) = 5$

$\frac{h_1}{h_0} = \frac{25}{5} = 5$

20. (b)  $\left(\frac{H_{\max}}{H_{\max}}\right)_1 = \frac{u^2 \sin^2(45^\circ - \alpha)}{u^2 \sin^2(45^\circ + \alpha)}$

$= \left(\frac{\frac{1}{2} \cos \alpha - \frac{1}{2} \sin \alpha}{\frac{1}{2} \cos \alpha + \frac{1}{2} \sin \alpha}\right)^2$

$= \frac{1 - \sin 2\alpha}{1 + \sin 2\alpha}$

21. (Bonus)

$\frac{u_1}{u_2} = \frac{45^\circ}{60^\circ}$

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22. (a) Maximum height  $H = \frac{v^2 \sin^2 \theta}{2g}$

$4R = \frac{4\pi^2 R^2}{T^2 2g} \sin^2 \theta \Rightarrow \left[\frac{2\pi R}{T} \sin \theta\right]^2 = \frac{4\pi^2 R^2}{T^2 2g} \sin^2 \theta$

$\Rightarrow \sin \theta = \sqrt{\frac{2gT^2}{\pi^2 R}}$

$\Rightarrow \theta = \sin^{-1} \left(\sqrt{\frac{2gT^2}{\pi^2 R}}\right)^2$

23. [16] For projectile maximum height,

$H_{\max} = \frac{u^2 \sin^2 \theta}{2g}$

$\frac{H_{\max}}{H_{\max}} = \frac{u_1^2}{u_2^2}$

$\frac{64}{H_{\max}} = \left(\frac{u}{2}\right)^2$

$H_{\max} = 16 \text{ m}$

24. (b) Here horizontal range and maximum height is same, so,

$\frac{u^2 \sin 2\theta}{g} = \frac{u^2 \sin^2 \theta}{2g}$

$4 \sin \theta \cos \theta = \sin^2 \theta$

$4 = \tan \theta$

25. (c) Range,

$R = \frac{u^2 \sin 2\theta}{g} \Rightarrow R \propto \sin(2\theta)$

$\frac{R_1}{R_2} = \frac{\sin(2\theta_1)}{\sin(2\theta_2)} = \frac{\sin(2 \times 15^\circ)}{\sin(2 \times 45^\circ)} = \frac{\sin 30^\circ}{\sin 90^\circ}$

$= \frac{1}{2}$

$R_2 = 2 R_1 = 2 \times 50 = 100 \text{ m}$

26. (b) Velocity of the projectile at the highest point =  $u \cos \theta$

$u \cos \theta = \frac{\sqrt{3}u}{2}$

$\Rightarrow \theta = 30^\circ$

27. (a)  $y = x - \frac{x^2}{20}$

For maximum height,

$\frac{dy}{dx} = 0 \Rightarrow \frac{d}{dx} \left(x - \frac{x^2}{20}\right) = 0$

$1 - \frac{2x}{20} = 0$

$x = 10$

So,  $y_{\max} = 10 - \frac{10^2}{20} = 5 \text{ m}$

28. (a) At the highest point  
Vertical component of the velocity,  $v_y = 0$   
Horizontal component of the velocity

$v_x = u_x = u \cos \theta$

Gravitational potential energy

$U_x = mgh$ , it is maximum at  $H_{\max}$

29. (b)  $R = \frac{u^2 \sin 2\theta}{g}$

$R$  is maximum for  $2\theta = 90^\circ$

$\sin 2\theta = \sin 90^\circ = 1$

$\therefore 2\theta = 90^\circ \Rightarrow \theta = 45^\circ$

30. (a) Horizontal component of velocity,  $v_x = u \cos 30^\circ = 40 \times \frac{\sqrt{3}}{2} = 20\sqrt{3} \text{ m/s}$

Vertical component of velocity,

$u_y = u \sin 30^\circ$

$= 40 \times \frac{1}{2} = 20 \text{ m/s}$

At  $t = 2 \text{ s}$ ,  $v_x = u_x = 20\sqrt{3}$

$v_y = u_y - gt = 20 - 10 \times 2 = 0$

$v = \sqrt{v_x^2 + v_y^2} = 20\sqrt{3} \text{ m/s}$

31. (d) Range =  $\frac{u^2 \sin 2\theta}{g}$

Range for angle of projection " $\alpha$ "

$R_1 = \frac{u^2 \sin 2\alpha}{g}$

Range for angle of projection " $\beta$ "

$R_2 = \frac{u^2 \sin 2\beta}{g}$

Given,  $\alpha + \beta = 90^\circ$

$\Rightarrow \beta = 90^\circ - \alpha$

$R_2 = \frac{u^2 \sin 2(90^\circ - \alpha)}{g}$

$= \frac{u^2 \sin 2\alpha}{g}$

$\Rightarrow \frac{R_1}{R_2} = \frac{\frac{u^2 \sin 2\alpha}{g}}{\frac{u^2 \sin 2\alpha}{g}} = 1$

32. (c) Maximum height

$H_{\max} = \frac{u^2 \sin^2 \theta}{2g} \Rightarrow H_{\max} = \sin^2 \theta$

$\frac{H_1}{H_2} = \frac{\sin^2 \theta_1}{\sin^2 \theta_2} = \frac{\sin^2 30^\circ}{\sin^2 60^\circ} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$

$\frac{H_1}{H_2} = \frac{\sin^2 \theta_1}{\sin^2 \theta_2} = \frac{\sin^2 30^\circ}{\sin^2 60^\circ} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$

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$\frac{H_1}{H_2} = \frac{\sin^2 \theta_1}{\sin^2 \theta_2} = \frac{\sin^2 30^\circ}{\sin^2 60^\circ} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$

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33. (c) Maximum vertical height is given by

$$H_{\max} = \frac{v^2}{2g} = 136 \text{ m}$$

and maximum horizontal range is

$$R_{\max} = \frac{v^2}{g} = 2H_{\max}$$

$$= 2(136)$$

$$\Rightarrow R_{\max} = 272 \text{ m}$$

34. [80] Time of flight  $T_f = 3 + 5 = 8 \text{ sec}$

Formula of Time of flight,

$$\therefore T = \frac{2u \sin \theta}{g}$$

$$u = \frac{Tg}{2 \sin 30^\circ}$$

$$= \frac{8 \times 10}{2 \times \left(\frac{1}{2}\right)}$$

$$u = 80 \text{ m/s}$$

35. [80] For the same horizontal range if angle of projection for projectile 1 is  $60^\circ$  then angle of projectile 2 is  $30^\circ$  for same range

$$H_1 = \frac{u^2 \sin^2 60^\circ}{2g}, H_2 = \frac{u^2 \sin^2 30^\circ}{2g}$$

$$H_1 + H_2 = \frac{u^2 \sin^2 60^\circ}{2g} + \frac{u^2 \sin^2 30^\circ}{2g}$$

$$= \frac{u^2}{2g} = \frac{40^2}{2 \times 10} = 80 \text{ m}$$

Hence, the sum of the maximum heights attained by the two projectile is 80 m.

36. (d) At the maximum height velocity =  $u \cos \theta = u \cos 30^\circ$

$$\frac{KE_{\text{initial}}}{KE_{\text{top}}} = \frac{\frac{1}{2} M(u^2)}{\frac{1}{2} M(u \cos 30^\circ)^2} = \frac{4}{3}$$

37. (d)  $R_1 = \frac{u^2 \sin 2\theta_1}{g}, R_2 = \frac{u^2 \sin 2\theta_2}{g}$

$$\frac{R_1}{R_2} = \frac{40^2 \sin(2 \times 30^\circ)}{60^2 \sin(2 \times 60^\circ)} = \frac{4 \sin 60^\circ}{9 \cos 30^\circ} = \frac{4}{9}$$

38. (a) For first angle,

$$\theta_1 = \theta, \text{ Range, } R = \frac{u^2 \sin 2\theta}{g}$$

Another angle  $(\theta_2 = 90^\circ - \theta)$  for which range will be same as that of  $\theta_1 = \theta$

$$R^1 = \frac{u^2 \sin(2(90^\circ - \theta))}{g} = \frac{u^2 \sin 2\theta}{g} = R$$

$$\text{at } \theta_1 = \theta, \theta_2 = \frac{u^2 \sin^2 \theta}{2g}$$

$$\& \theta_2 = 90^\circ - \theta, \theta_2 = \frac{u^2 \sin^2 \theta_2}{2g} = \frac{u^2 \cos^2 \theta}{2g}$$

$$\Rightarrow h_1 h_2 = \left[ \frac{u^2 \sin \theta \cos \theta}{2g} \right]^2 = \left[ \frac{u^2 \sin 2\theta}{4g} \right]^2$$

$$\Rightarrow h_1 h_2 = \frac{1}{4} \left[ \frac{u^2 \sin 2\theta}{g} \right]^2$$

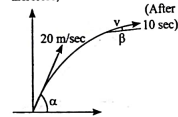
$$\Rightarrow h_1 h_2 = \frac{1}{4} \left[ \frac{u^2 \sin 2\theta}{g} \right]^2$$

$$h_1 h_2 = \frac{1}{4} R^2$$

$$R = 4\sqrt{h_1 h_2}$$

Hence, Both statement is true & Reason is correct explanation for assertion.

39. (b) Throughout the motion, the horizontal speed will remain constant, therefore,



Along x-axis,

$$u_x = v_x = 20 \cos \alpha$$

Along y-axis

$$u_y = 20 \sin \alpha,$$

Applying Kinematic eqn.

$$v_y = u_y + at$$

$$v_y = 20 \sin \alpha - 10 \times 10$$

$$\tan \beta = \frac{v_y}{v_x} = \frac{20 \sin \alpha - 100}{20 \cos \alpha}$$

$$\Rightarrow \tan \beta = \tan \alpha - 5 \sec \alpha$$

40. (c) Range,  $R = \frac{u^2 \sin 2\theta}{g}$

$$R_1 = \frac{u^2 \sin 90^\circ}{g}$$

$$R_2 = \frac{u^2 \sin 60^\circ}{g}$$

$$\Rightarrow \text{Required ratio, } \frac{R_1}{R_2} = \frac{2}{\sqrt{3}}$$

41. (d)  $y = x - \frac{10x^2}{2u^2 \left(\frac{1}{2}\right)}$

$$\Rightarrow 10 = 20 - \frac{(10)(400)}{u^2} \Rightarrow u = 20 \text{ m/s}$$

$$t = \frac{T}{\sqrt{2}} = \frac{(2)(20) \sin 45^\circ}{\sqrt{2}(10)} = 2 \text{ s}$$

$$\vec{v} = u \cos \theta \hat{i} + (u \sin \theta - gt) \hat{j}$$

$$= 10\sqrt{2} \hat{i} + (10\sqrt{2} - 10(2)) \hat{j}$$

Momentum

$$\vec{p} = M\vec{v} = 100\sqrt{2} \hat{i} + (100\sqrt{2} - 200) \hat{j}$$

42. (d)  $R = \frac{v^2 (2 \sin \theta \cos \theta)}{g}$

$$t = \frac{V \sin \theta}{g} \Rightarrow V = \frac{gt}{\sin \theta}$$

$$\Rightarrow R = \frac{g^2 t^2}{\sin^2 \theta} \cdot \frac{2 \sin \theta \cos \theta}{g}$$

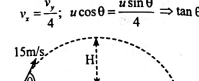
$$\tan \theta = \frac{2gt^2}{R} = \frac{20t^2}{R}$$

$$\cot \theta = \frac{R}{20t^2}$$

43. (d)  $R = H$

$$\frac{2v_x \times v_y}{g} = \frac{v_y^2}{2g}$$

$$v_x \times v_y = \frac{v_y^2}{2} \Rightarrow u \cos \theta = \frac{u \sin \theta}{2} \Rightarrow \tan \theta = 4$$



44. (b) According to question, we can write

$$R = \frac{u^2 \sin 2\alpha}{g}$$

$$\text{For maximum range, } (R_{\max})$$

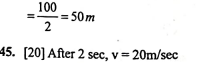
$$\theta = 45^\circ$$

$$R_{\max} = \frac{u^2}{g} \Rightarrow u^2 = g R_{\max}$$

$$\Rightarrow H_{\max} = \frac{u^2}{2g} = \frac{g R_{\max}}{2g} = \frac{R_{\max}}{2}$$

$$= \frac{100}{2} = 50 \text{ m}$$

45. [20] After 2 sec,  $v = 20 \text{ m/sec}$



$$\vec{v} = u \cos 45^\circ \hat{i} + (u \sin 45^\circ - gt) \hat{j}$$

$$\Rightarrow \vec{v} = u \cos 45^\circ \hat{i} + (u \sin 45^\circ - 20) \hat{j}$$

$$v = 20 = \sqrt{(u \cos 45^\circ)^2 + (u \sin 45^\circ - 20)^2}$$

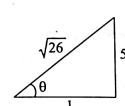
$$\Rightarrow 400 = u^2 + 400 - 40u \sin 45^\circ$$

$$\Rightarrow u = 40 \sin 45^\circ$$

$$H_{\max} = \frac{u^2 \sin^2 45^\circ}{2g} = \frac{40^2 \sin^2 45^\circ \sin^2 45^\circ}{2g} = 20 \text{ m}$$

46. [5] Given,  $u_x = \hat{i}, |u_x| = 1$

$$y = x5(1-x) = x \tan \theta \left(1 - \frac{x}{R}\right)$$



$$\tan \theta = \frac{5}{R}, R = 1$$

$$\tan \theta = \left| \frac{u_y}{u_x} \right|$$

$$|u_y| = |u_x| \tan \theta = 1 \times 5 = 5 \text{ m/s}$$

47. [15or75]  $R = \frac{u^2 \sin(2 \times 45^\circ)}{g} = \frac{u^2}{g}$

$$\frac{R}{2} = \frac{u^2}{2g} = \frac{u^2 \sin 2\theta}{2g}$$

$$\Rightarrow \sin 2\theta = \frac{1}{2} \Rightarrow \theta = 15^\circ, 75^\circ$$

48. (c) At highest point, velocity

$$V = u \cos 60^\circ = \frac{u}{2}$$

$$\text{K.E at topmost point} = \frac{1}{2} m \left( \frac{u}{2} \right)^2 = \frac{E}{4}$$

49. (c) Time taken to reach maximum height

$$t = \frac{u \sin \theta}{g} \therefore \frac{u_1 \sin \theta_1}{g} = \frac{u_2 \sin \theta_2}{g}$$

$$\Rightarrow u_1 \sin 30^\circ = u_2 \sin 45^\circ$$

$$\Rightarrow \frac{u_1}{u_2} = \frac{1/\sqrt{2}}{1/2} = \frac{\sqrt{2}}{1}$$

50. [1] Time of flight is same

$$\Rightarrow \text{Vertical component of velocity is same} \Rightarrow H_{\max} \text{ is same}$$

51. (b)

$$5\sqrt{3} \text{ m/s}$$

$$10 \text{ m/s}$$

$$5 \text{ m/s}$$

$$60^\circ$$

$$\alpha$$

$$\frac{1}{2}g = 10$$

$$\text{At } t = 1$$

$$u_x = 5, u_y = 5\sqrt{3}$$

$$v_x = 5\sqrt{3} - 10; v_y = 5$$

Angle made by velocity vector at  $t = 1$  sec.

$$|\tan \alpha| = \left| \frac{v_y}{v_x} \right| = \left| \frac{5}{5\sqrt{3} - 10} \right| = 2 - \sqrt{3}$$

$$\alpha = 15^\circ$$

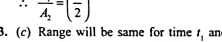
$$\therefore R = \frac{5^2 + (10 - 5\sqrt{3})^2}{10 \cos \alpha} = \frac{200 - 100\sqrt{3}}{10 \times 0.965}$$

$$= 2.8 \text{ m}$$

52. (a) Area  $\propto \pi (\text{Range})^2 \propto u^4$

$$\therefore \frac{A_1}{A_2} = \left( \frac{u_1}{u_2} \right)^4$$

53. (c) Range will be same for time  $t_1$  and  $t_2$ , so angles of projection will be  $\theta$  and  $90^\circ - \theta$



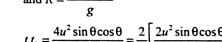
$$t_1 = \frac{2u \sin \theta}{g}, t_2 = \frac{2u \sin(90^\circ - \theta)}{g}$$

$$\text{and } R = \frac{u^2 \sin 2\theta}{g}$$

$$t_1 t_2 = \frac{4u^2 \sin \theta \cos \theta}{g^2} = \frac{2}{g} \left[ \frac{2u^2 \sin \theta \cos \theta}{g} \right]$$

$$= \frac{2R}{g}$$

54. (b)



$$\text{For same range angle of projection will be } \theta \text{ and } 90^\circ - \theta$$

$$R = \frac{u^2 \sin 2\theta \cos \theta}{g}$$

$$R = \frac{u^2 \sin^2 \theta}{2g}$$

$$h_2 = \frac{u^2 \sin^2(90^\circ - \theta)}{2g} \therefore \frac{R^2}{h_1 h_2} = 16$$

55. (d) Equation of trajectory is given as

$$y = 2x - 9x^2$$

Comparing with equation:

$$y = x \tan \theta - \frac{g}{2u^2 \cos^2 \theta} x^2$$

$$\text{We get; } \tan \theta = 2$$

$$\therefore \cos \theta = \frac{1}{\sqrt{5}}$$

$$\text{Also, } \frac{R}{2u^2 \cos^2 \theta} = 9$$

$$\Rightarrow \frac{10}{2 \times 9 \times \left( \frac{1}{\sqrt{5}} \right)^2} = u^2$$

$$\Rightarrow u^2 = \frac{25}{9}$$

$$\Rightarrow u = \frac{5}{3} \text{ m/s}$$

56. (d) Horizontal speed  $u = 360 \text{ km/h}$

$$= 100 \text{ m/s}$$

$$\text{Height } H = 2 \text{ km} = 2000 \text{ m}$$

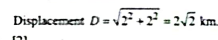
$$\text{Fall time } t = \sqrt{\frac{2H}{g}} = \sqrt{\frac{4000}{10}} = 20 \text{ s}$$

$$\text{Horizontal distance } x = u t$$

$$= (100 \times 20) = 2 \text{ km}$$

$$\text{Displacement } D = \sqrt{2^2 + 2^2} = 2\sqrt{2} \text{ km}$$

57. [2]



Just middle (for minimum speed)

The ball needs to just cross 4 steps to just hit 5th step

Therefore, horizontal range (R) = 0.4 m

$$R = u t$$

Similarly, in vertical direction

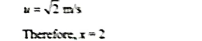
$$h = \frac{1}{2} g t^2$$

$$0.4 = \frac{1}{2} g \left( \frac{0.4}{u} \right)^2$$

$$u = \sqrt{2} \text{ m/s}$$

$$\text{Therefore, } x = 2$$

58. [100] H is the height at which the object is thrown



100m

x

this is the case of horizontal projectile

$$100 = \sqrt{\frac{2H}{g}} \cdot x = \frac{v}{2} \sqrt{\frac{2(4H)}{g}} = \frac{v}{2} \sqrt{\frac{2H}{g}}$$

$$\Rightarrow x = 100$$

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$$84. [200] \quad \omega^2 = \omega_0^2 + 2\alpha\theta \Rightarrow \left(\frac{2\pi}{60} \times 1800\right)^2$$

$$= \left(\frac{2\pi}{60} \times 600\right)^2 + 2\left(\frac{2\pi}{60} \times \frac{1200}{10}\right)\theta$$

$$\text{On solving, } \theta = \frac{80\pi \times 40\pi}{2 \times 4\pi} = 400\pi \text{ rad}$$

$$\text{Number of rotations} = \frac{\theta}{2\pi} = \frac{400\pi}{2\pi} = 200$$

$$85. (a) \quad \vec{r} = \cos \omega t \hat{i} + \sin \omega t \hat{j}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \omega(-\sin \omega t \hat{i} + \cos \omega t \hat{j})$$

$$\vec{a} = \frac{d\vec{v}}{dt} = -\omega^2(\cos \omega t \hat{i} + \sin \omega t \hat{j})$$

$$\vec{a} = -\omega^2 \vec{r}$$

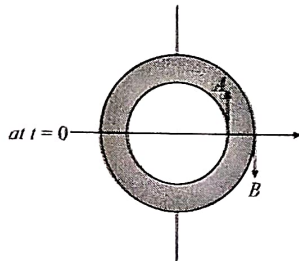
$\therefore \vec{a}$  is antiparallel to  $\vec{r}$

$$\vec{v} \cdot \vec{r} = \omega(-\sin \omega t \cos \omega t + \cos \omega t \sin \omega t) = 0$$

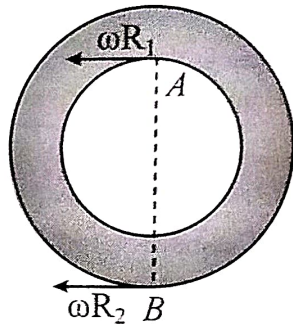
so  $\vec{v} \perp \vec{r}$

$$86. (d) \quad \Delta \vec{v} = 2v \sin\left(\frac{\theta}{2}\right)$$

$$87. (c)$$



$$\text{at } t = \frac{\pi}{2\omega}$$



$$\vec{v}_A - \vec{v}_B = -\omega R_1 \hat{i} + \omega R_2 \hat{i}$$

$$= \omega(R_2 - R_1) \hat{i}$$